EQUAL MAPS IN THE DERIVED CATEGORY BETWEEN FINITE COMPLEXES OF FINITE LOCALLY FREE MODULES BECOME HOMOTOPIC AFTER A PERFECT RESOLUTION OF THE SOURCE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file perfect.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- A quasi-compact quasi-separated scheme with the resolution property has affine diagonal. invoked at proof lines 9966-9969. Unlocks the quasi-isomorphism existence proposition.
- For a scheme with affine diagonal, any map in the derived category between complexes of quasi-coherent modules is represented by an actual map of complexes after pullback along a quasi-isomorphism, and equal derived maps become homotopic maps of complexes. invoked at proof lines 9970-9977. Produces the initial resolution G^bullet with homotopy h witnessing the equality.
- Any bounded complex of quasi-coherent modules on a scheme with the resolution property receives a quasi-isomorphism from a bounded complex of finite locally free modules. invoked at proof lines 9985-9987. Replaces the truncated quasi-coherent resolution by a perfect complex, finishing the proof.

Source reference. Original lemma 0F8C: sources/stacks-perfect/pinned-original-source.tex, source lines 9705–9705. Cited at proof lines 9966-9968 to conclude X has affine diagonal; the application of the diagonal proposition is inside the counted span.

Source reference. Original proposition 08DB: sources/stacks-perfect/pinned-original-source.tex, source lines 1801–1801. Cited at proof lines 9969-9977 to produce the initial resolution G^bullet with homotopy; the truncation and replacement steps are inside the counted span.

Source reference. Original lemma 0F8E: sources/stacks-perfect/pinned-original-source.tex, source lines 9790–9790. Cited at proof lines 9985-9987 to finish by replacing the truncated resolution; the composition producing the final gamma^bullet is inside the counted span.

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Lemma 0.2. *Let X be a quasi-compact and quasi-separated scheme with the resolution property. Let $\mathcal{E}^\bullet$ and $\mathcal{F}^\bullet$ be finite complexes of finite locally free $\mathcal{O}_X$-modules. Let $\alpha^\bullet, \beta^\bullet : \mathcal{E}^\bullet \to \mathcal{F}^\bullet$ be two maps of complexes defining the same map in $D(\mathcal{O}_X)$. Then there exists a quasi-isomorphism $\gamma^\bullet : \mathcal{G}^\bullet \to \mathcal{E}^\bullet$ where $\mathcal{G}^\bullet$ is a bounded complex of finite locally free $\mathcal{O}_X$-modules such that $\alpha^\bullet \circ \gamma^\bullet$ and $\beta^\bullet \circ \gamma^\bullet$ are homotopic maps of complexes.*

Proof. By Lemma 0F8C we see that X has affine diagonal. Hence by Proposition 08DB (and the definition of the derived category) there exists a quasi-isomorphism $\gamma^\bullet : \mathcal{G}^\bullet \to \mathcal{E}^\bullet$ where $\mathcal{G}^\bullet$ is a complex of quasi-coherent $\mathcal{O}_X$-modules such that $\alpha^\bullet \circ \gamma^\bullet$ and $\beta^\bullet \circ \gamma^\bullet$ are homotopic maps of complexes. Choose a homotopy $h^i : \mathcal{G}^i \to \mathcal{F}^{i-1}$ witnessing this fact. Choose $n \ll 0$. Then the map $\gamma^\bullet$ factors canonically over the quotient map $\mathcal{G}^\bullet \to \tau_{\geq n}\mathcal{G}^\bullet$ as $\mathcal{E}^\bullet$ is bounded below. For the exact same reason the maps h^i will factor over the surjections $\mathcal{G}^i \to (\tau_{\geq n}\mathcal{G})^i$. Hence we see that we may replace $\mathcal{G}^\bullet$ by $\tau_{\geq n}\mathcal{G}^\bullet$. Then we may apply Lemma 0F8E to conclude. □